Midterm 2 Review

CSCI 1101

2024-04-05

About the Exam

Specifics:

- In class next Tuesday.
- Paper-and-pencil based.
- You may bring one index card (or $1/2$ sheet of notebook paper) with notes.
- Covers material though week 10.

Resources:

- lecture notes, homeworks, in-class activities
- practice exam on canvas
- CodeRunner Exam Practice (up to Mixed Practice 2)

Mutation

What does it mean for data to be mutable?

Are the following python types mutable or immutable?

- `bool`
- `str`
- `list`
- `tuple`
- `set`
- `dict`

For each mutable type above, give an example of a function, method, or operator that can *change* the value of its elements.

Iteration

What does it mean for a Python type to be *iterable?*

Give some examples of iterable and non-iterable types.

Data Structures

Python includes some built-in data structures that we can use to contain and organize the data in our programs.

Describe each of the following data structures and give an example of their use:

- lists
- tuples
- sets
- dictionaries

Activity: Sets

Recall that *sets* have a theory that lets us reason about them abstractly.

Consider the following sets:

```
(A , B , C) = ({1,2,3,4} , {2,4,6,8} , {3,4,5,6})
```

For each of the following expressions:

- draw a Venn diagram of the sets involved,
- translate the expression into Python's set notation (method or operator),
- predict the value of the expression, then check your answer by evaluating it.

1. $\mathrm{A} \cap \mathrm{B}$
2. $(\mathrm{A} \cup \mathrm{B}) - \mathrm{C}$
3. $\mathrm{A} \cap \mathrm{B} \cap \mathrm{C} \subseteq \mathrm{A}$

Activity: Tabular Data

Recall that we can use a *list* of *tuples* to represent a table.

```
# signature:  list (tuple [str, int])
beatle_grades = \
[   #  name    , grade
    ("John"    , 94) ,
    ("Paul"    , 82) ,
    ("George"  , 79) ,
    ("Ringo"   , 66) ]
```

Write a function that returns the name of the student with the highest grade:
`best_student : (list (tuple (str , int))) --> str`.

For example:

```
> best_student(beatle_grades)
'John'
```

Activity: Mixed Data Structures

Suppose that the global variable `prices : dict (str, float)` contains the prices of fruits for sale:

```
prices = \
  {'apple':0.50, 'banana':0.25, 'kiwi':1.00, 'lime':0.75}
```

We can represent an *order* as a `list (tuple [str, int])`, for example:

```
my_order = [('banana', 2) , ('kiwi', 1) , ('lime', 1)]
```

Write a function `fruit_bill : (list (tuple [str,int])) --> float` that calculates the total price of the fruits in an order. For example:

```
> fruit_bill(my_order)
2.25
```

To Do

Study for Midterm 2 in class next Tuesday, April 9.

Finish *Homework 9: Exceptions* on CodeRunner by 11:00AM next Thursday.

Finish on *Project 3 - Automated Grader (Part 2)* on CodeRunner by Friday, April 19.